

University of Engineering and Technology Lahore

— Course Outline Report —

CSC201L — Information Security Lab

WEEK 1

Lab Manual

Introduction to Lab Environment:
Linux Terminal and Virtual Lab Setup

CLO Addressed: CLO1 — Demonstrate basic cryptographic processes
Duration: 3 Hours
Tools Required: VMware Workstation Pro, Kali Linux
Prerequisites: Basic computer literacy

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1 BRIEF THEORY

1.1 What is Information Security?

Information Security is the practice of protecting information from unauthorized access, modification, or destruction. This lab course will teach you practical security skills using real tools and environments.

Before we dive into security topics (starting Week 2), we need to first set up our **lab environment** — the tools and systems we will use throughout the entire semester.

1.2 Why Linux?

Throughout this course, we will work primarily in Linux. Here is why:

Reason	Explanation
Server Dominance	Over 90% of the world's servers run Linux
Security Tools	Most security tools (Nmap, Wireshark, Metasploit) are built for Linux
Open Source	Source code is available for inspection — no hidden backdoors
Command Line Power	Terminal gives precise, scriptable control over the entire system
Industry Standard	Security professionals use Linux daily
Free	No licensing costs

Table 1: Reasons for Linux dominance in cybersecurity

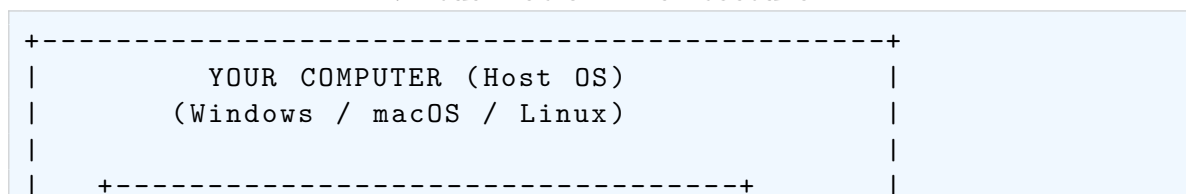
1.2.1 What is Kali Linux?

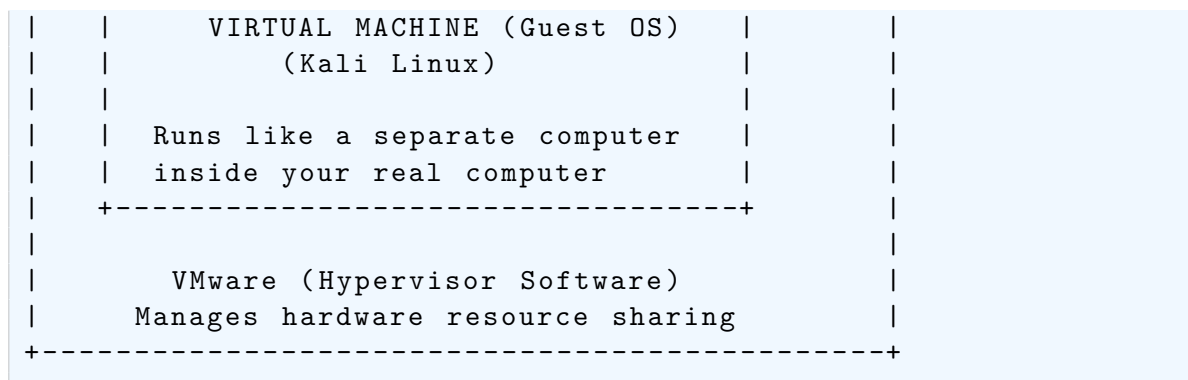
Kali Linux is a **Debian-based Linux distribution** built specifically for cybersecurity. It comes pre-installed with **600+ security tools** for penetration testing, forensics, and security research. We will use Kali Linux as our primary lab operating system.

1.3 Why Virtual Machines?

We will run Kali Linux inside a **Virtual Machine (VM)** rather than installing it directly on your computer.

Virtualization Architecture





Benefit	Why It Matters
Isolation	Experiments stay inside the VM — your real computer is safe
Snapshots	Save VM state and restore instantly if something breaks
Safe Testing	Test attacks, malware, exploits without any real risk
Same Environment	Every student gets an identical setup

Table 2: Benefits of using virtual machines in a security lab

1.4 Linux Terminal — What You Need to Know

The **terminal** is a text-based interface where you type commands. Unlike clicking icons, the terminal gives you direct control over the system.

1.4.1 The Terminal Prompt

```
+---(kali@kali)-[~]
+--$ _

kali      = your username
kali      = computer name (hostname)
~         = current directory (~ means /home/kali)
$         = regular user (# means root/admin)
```

1.4.2 How Commands Work

```
command [options] [arguments]

Example: ls -la /home/kali
          |   |   |
          |   |   +-- where to look (argument)
          |   +----- how to display (-l long, -a all)
          +----- what to do (list files)
```

That is all the theory you need. **Let's get hands-on.**

2 ENVIRONMENT SETUP (VMware)

2.1 Download VMware Workstation Pro

Note

VMware Workstation Pro is **FREE** for personal use since November 2024.

1. Go to: <https://support.broadcom.com/>
2. Click **"Register"** -> Create a **free** Broadcom account
3. Use your university email
4. Verify your email by clicking the link **in** your inbox
5. Log **in** to your account
6. Search **for** **"VMware Workstation Pro"**
7. Download the latest version:
+-----+
| Windows: VMware-workstation-full-xx.exe |
| Linux: VMware-Workstation-xx.bundle |
| macOS: Download VMware Fusion instead |
+-----+
8. Save the installer to your Downloads folder

2.2 Install VMware

2.2.1 Windows Installation

- Step 1: Right-click installer -> **"Run as administrator"**
- Step 2: Click **"Next"** on welcome screen
- Step 3: Accept license agreement -> **"Next"**
- Step 4: Keep default install path
[x] Add VMware console tools to system PATH
[x] Install Enhanced Keyboard Driver
-> **"Next"**
- Step 5: Uncheck update/CEIP options -> **"Next"**
- Step 6: [x] Desktop Shortcut -> **"Next"**
- Step 7: Click **"Install"** -> Wait 3-5 minutes
- Step 8: Click **"Finish"**
- Step 9: Restart your computer

2.2.2 macOS Installation (VMware Fusion)

- Step 1: Download VMware Fusion from Broadcom portal
- Step 2: Double-click .dmg -> Double-click installer
- Step 3: Enter macOS admin password
- Step 4: Accept license -> Select **"Personal Use (Free)"**
- Step 5: Grant permissions **in** System Preferences -> Privacy
- Step 6: Restart **if** prompted

2.2.3 Linux Installation

```
chmod +x VMware-Workstation-Full-17.x.x-xxxxx.x86_64.bundle
sudo ./VMware-Workstation-Full-17.x.x-xxxxx.x86_64.bundle
# Follow prompts, accept license, keep defaults
```

2.3 First Launch of VMware

```
Step 1: Double-click VMware Workstation icon
Step 2: Select: (*) "Use VMware Workstation for Personal Use"
Step 3: Click "Continue" -> "Finish"
Step 4: VMware home screen appears -- ready to use
```

2.4 Download Kali Linux for VMware

1. Go to: <https://www.kali.org/get-kali/#kali-virtual-machines>
2. Click the "VMware" tab (NOT VirtualBox)
3. Download the 64-bit VMware image
File: kali-linux-2024.x-vmware-amd64.7z (~3-4 GB)
4. Save to your Downloads folder

2.4.1 Extract the Downloaded File

The file is compressed in .7z format. You need to extract it:

Windows:

1. Download 7-Zip from: <https://www.7-zip.org/> (if not installed)
2. Right-click the .7z file -> 7-Zip -> "Extract Here"
3. Wait for extraction to complete
4. Move the extracted folder to: C:\VMs\Kali-Linux\
WARNING: Do NOT leave it in Downloads

macOS:

1. Install "The Unarchiver" from App Store
2. Double-click the .7z file to extract
3. Move extracted folder to: ~/Virtual Machines/Kali-Linux/

Linux:

```
sudo apt install p7zip-full
7z x kali-linux-2024.x-vmware-amd64.7z
mkdir -p ~/VMs
mv kali-linux-* ~/VMs/Kali-Linux/
```

2.5 Import Kali Linux into VMware

```
Step 1:  Open VMware Workstation
Step 2:  Click "Open a Virtual Machine" (or File -> Open)
Step 3:  Navigate to the extracted Kali folder
Step 4:  Select the .vmx file -> Click "Open"
Step 5:  If asked "Did you copy or move this VM?"
        -> Select "I Copied It"
Step 6:  The VM now appears in your library
```

2.5.1 Configure VM Settings (Before First Boot)

```
Step 1:  Right-click the VM -> "Settings..."

Step 2:  Memory: Set to 4096 MB
        (minimum 2048 MB if your system has limited RAM)

Step 3:  Processors: Set to 2 cores

Step 4:  Network Adapter:
        (*) NAT
        [x] Connect at power on

Step 5:  Click "OK"
```

2.6 First Boot

```
Step 1:  Select Kali Linux VM -> Click green "Play" Power on
Step 2:  Wait for boot -> Login screen appears
Step 3:  Login:
        +-----+
        | Username:  kali          |
        | Password:  kali          |
        +-----+
```

2.6.1 Install VMware Tools (Do This Immediately)

Open terminal (Ctrl + Alt + T) and run:

```
sudo apt update
sudo apt install -y open-vm-tools open-vm-tools-desktop
sudo reboot
```

After reboot, you will have: screen auto-resize, copy-paste between host and VM, drag-and-drop file transfer.

2.6.2 Change Default Password

```
passwd
# Current password: kali
# New password: [choose something strong]
# Confirm: [same password]
```

Warning

Remember your new password! Write it down somewhere safe. If you forget it, you will need to restore from a snapshot.

2.6.3 Take Your First Snapshot

```
VM -> Snapshot -> Take Snapshot
Name: "Week01_FreshSetup"
Description: "Clean Kali, VMware Tools installed, password
            changed"
Click OK
```

Tip

What is a snapshot? It saves the exact state of your VM. If you break something later, you can instantly restore to this clean state.

2.7 Online Lab Alternatives

If you **cannot install VMware** on your computer (hardware limitations, no admin access, etc.), you can use these online alternatives:

2.7.1 Option A: TryHackMe (Recommended Online Alternative)

1. Go to: <https://tryhackme.com/>
2. Create a **free** account
3. Use their browser-based **"AttackBox"** -- a full Kali Linux environment running **in** your browser
4. No installation needed
5. Free tier gives limited hours per day

2.7.2 Option B: Hack The Box

1. Go to: <https://www.hackthebox.com/>
2. Create a **free** account
3. Use their **"Pwnbox"** -- browser-based Parrot OS environment
4. Similar to Kali Linux with security tools pre-installed

2.7.3 Option C: Google Cloud Shell

1. Go to: <https://shell.cloud.google.com/>
2. Log **in** with a Google account
3. Provides a **free** Linux terminal (Debian-based)
4. Limited security tools but good **for** basic Linux practice
5. No VM setup needed

2.7.4 Option D: JSLinux (No Account Needed)

1. Go to: <https://bellard.org/jslinux/>
2. Select "**Alpine Linux**" or "**Linux Console**"
3. A Linux terminal runs directly **in** your browser
4. Very basic -- good **for** practicing commands only

2.7.5 Platform Comparison

Platform	Full Kali?	Browser?	Free?	Best For
VMware (Local)	Yes	No	Yes	Full lab experience (recommended)
TryHackMe	Yes	Yes	Limited	Students without powerful PCs
Hack The Box	Parrot	Yes	Limited	Alternative to TryHackMe
Cloud Shell	No	Yes	Yes	Basic Linux practice
JSLinux	No	Yes	Yes	Quick terminal practice

Table 3: Comparison of lab environment options

Note

Instructor recommendation: Use VMware with a local VM if at all possible. Online labs have time limits and may not have all the tools we need in future weeks.

3 INSTRUCTOR DEMONSTRATIONS

Students: Watch carefully, follow along on your machine, and take notes.

Demo 1: Your First Commands

Instructor opens terminal and types:

```
whoami
```

```
kali
```

```
hostname
```

```
kali
```

```
pwd
```

```
/home/kali
```

Instructor explains: These three commands answer the most basic questions: Who am I? What computer am I on? Where am I in the file system?

```
uname -a
```

```
Linux kali 6.1.0-kali5-amd64 #1 SMP PREEMPT_DYNAMIC Debian  
6.1.12-1kali2 x86_64 GNU/Linux
```

```
id
```

```
uid=1000(kali) gid=1000(kali) groups=1000(kali),24(cdrom),25(  
floppy)...
```

```
date
```

```
Wed Jan 15 10:30:00 EST 2025
```

```
cal
```

```
      January 2025  
Su Mo Tu We Th Fr Sa  
      1  2  3  4  
5   6  7  8  9 10 11
```

```
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31
```

```
uptime
```

```
10:30:00 up 25 min, 1 user, load average: 0.15, 0.10, 0.05
```

Demo 2: Navigating the File System

```
# Where am I?
pwd
```

```
/home/kali
```

```
# What is here?
ls
```

```
Desktop  Documents  Downloads  Music  Pictures  Public
Templates  Videos
```

```
# Show details (permissions, size, date)
ls -l
```

```
drwxr-xr-x 2 kali kali 4096 Jan 10 08:00 Desktop
drwxr-xr-x 2 kali kali 4096 Jan 10 08:00 Documents
drwxr-xr-x 2 kali kali 4096 Jan 10 08:00 Downloads
...
```

```
# Show hidden files too
ls -la
```

```
drwxr-xr-x 15 kali kali 4096 Jan 15 10:00 .
drwxr-xr-x  3 root root 4096 Jan 10 08:00 ..
-rw-----  1 kali kali  220 Jan 10 08:00 .bash_logout
-rw-----  1 kali kali 3526 Jan 10 08:00 .bashrc
-rw-----  1 kali kali  807 Jan 10 08:00 .profile
drwxr-xr-x  2 kali kali 4096 Jan 10 08:00 Desktop
...
```

Instructor explains:

- d at the start = directory, - = regular file
- . = current directory, .. = parent directory

- Files starting with `.` are hidden (e.g., `.bashrc`)

```
# Go to another directory
cd /etc
pwd
ls | head -10
```

```
/etc
adduser.conf
alternatives
apt
bash.bashrc
ca-certificates
cron.d
cron.daily
dbus-1
default
deluser.conf
```

```
# Go deeper
cd /var/log
pwd
ls | head -10
```

```
/var/log
alternatives.log
apt
auth.log
boot.log
btmpt
...
```

```
# Go back home (three different ways)
cd ~
# OR
cd /home/kali
# OR just
cd

pwd
```

```
/home/kali
```

```
# Go to parent directory
cd ..
pwd
```

```
/home
```

```
cd ~
```

Instructor shows the file system map:

```
/ (root -- top of everything)
|-- /home/kali    <-- You are usually here
|-- /etc          <-- Configuration files
|-- /var/log      <-- Log files
|-- /bin          <-- Basic commands (ls, cp, cat)
|-- /tmp          <-- Temporary files
|-- /usr          <-- User programs
+-- /root         <-- Root user's home (NOT the same as /)
```

Demo 3: Creating Directories and Files

```
cd ~
```

```
# Create a directory
mkdir mylab
ls
```

```
Desktop  Documents  Downloads  Music  mylab  Pictures  ...
```

```
# Go into it
cd mylab

# Create nested directories in one command
mkdir -p project/src/docs
ls -R
```

```
.:
project

./project:
src

./project/src:
docs

./project/src/docs:
```

```
# Create an empty file
touch notes.txt
ls -la notes.txt
```

```
-rw-r--r-- 1 kali kali 0 Jan 15 10:40 notes.txt
```

```
# Write text into a file (> creates or overwrites)  
echo "This is my first file" > notes.txt  
cat notes.txt
```

```
This is my first file
```

```
# Append text to a file (>> adds to the end)  
echo "This is the second line" >> notes.txt  
cat notes.txt
```

```
This is my first file  
This is the second line
```

```
# Write multiple lines  
echo "Line 1" > data.txt  
echo "Line 2" >> data.txt  
echo "Line 3" >> data.txt  
cat data.txt
```

```
Line 1  
Line 2  
Line 3
```

```
# Copy a file  
cp notes.txt notes_backup.txt  
ls
```

```
data.txt  notes.txt  notes_backup.txt  project
```

```
# Rename (move) a file  
mv data.txt mydata.txt  
ls
```

```
mydata.txt  notes.txt  notes_backup.txt  project
```

```
# Delete a file  
rm notes_backup.txt  
ls
```

```
mydata.txt  notes.txt  project
```

```
# Count lines, words, characters
wc notes.txt
```

```
2  9  45 notes.txt
```

(2 lines, 9 words, 45 characters)

```
# View first/last lines of a file
head -1 notes.txt
tail -1 notes.txt
```

```
This is my first file
This is the second line
```

```
# Search for text inside a file
grep "second" notes.txt
```

```
This is the second line
```

```
# Create a file using nano editor
nano hello.txt
# Type some text -> Ctrl+X -> Y -> Enter to save
```

Demo 4: System Information Commands

```
# Disk space
df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/sda1	39G	12G	26G	32%	/
tmpfs	2.0G	0	2.0G	0%	/dev/shm

```
# Memory (RAM)
free -h
```

	total	used	free
Mem:	3.8Gi	1.2Gi	2.1Gi
Swap:	975Mi	0B	975Mi

```
# Running processes (press q to quit)
top
```

```
# Simpler process listing
```

```
ps aux | head -5
```

```
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.0   0.3 16743 12148 ?        Ss   10:00   0:01 /sbin/
  init
root         2   0.0   0.0      0     0 ?         S    10:00   0:00 [kthreadd
  ]
...
```

```
# Clear the terminal screen
```

```
clear
```

```
# See your command history
```

```
history | tail -20
```

Demo 5: Network Basics

```
# Your IP address
```

```
ip addr show
```

```
1: lo: <LOOPBACK,UP,LOWER_UP>
   inet 127.0.0.1/8 scope host lo
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP>
   inet 192.168.72.128/24 brd 192.168.72.255 scope global
   dynamic ens33
```

Instructor explains:

- lo = loopback (127.0.0.1 — machine talking to itself)
- ens33 = your network adapter (VMware uses ens33 typically)
- 192.168.72.128 = your VM's IP address

```
# Shorter way to see just IPs
```

```
ip -4 addr show | grep inet
```

```
inet 127.0.0.1/8 scope host lo
inet 192.168.72.128/24 scope global dynamic ens33
```

```
# Test internet connection
```

```
ping -c 3 google.com
```

```
PING google.com (142.250.185.78) 56(84) bytes of data.
64 bytes from 142.250.185.78: icmp_seq=1 ttl=128 time=15.2 ms
64 bytes from 142.250.185.78: icmp_seq=2 ttl=128 time=14.8 ms
64 bytes from 142.250.185.78: icmp_seq=3 ttl=128 time=15.1 ms
```

```
--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss
```

```
# Check open ports on your system
```

```
ss -tuln
```

```
# DNS lookup
```

```
nslookup google.com
```

```
Server:      192.168.72.2
Address:     192.168.72.2#53
```

```
Non-authoritative answer:
```

```
Name:       google.com
```

```
Address:    142.250.185.78
```

Demo 6: File Permissions

```
cd ~/mylab
ls -la notes.txt
```

```
-rw-r--r-- 1 kali kali 45 Jan 15 10:40 notes.txt
```

Instructor draws on board:

```
- rw- r-- r--
|  |  |  |
|  |  |  +-- Others: read only
|  |  +----- Group: read only
|  +----- Owner: read + write
+----- Type: regular file

r = read (4)      w = write (2)      x = execute (1)      - = none
(0)
```

```
# Make file private (owner only)
```

```
chmod 600 notes.txt
```

```
ls -la notes.txt
```

```
-rw----- 1 kali kali 45 Jan 15 10:40 notes.txt
```

```
# Remove ALL permissions
```

```
chmod 000 notes.txt
```

```
cat notes.txt
```

```
cat: notes.txt: Permission denied
```

```
# Restore
chmod 644 notes.txt
cat notes.txt
```

```
This is my first file
This is the second line
```

```
# Execute permission demo
echo '#!/bin/bash' > test.sh
echo 'echo "Script executed successfully!"' >> test.sh

./test.sh
```

```
bash: ./test.sh: Permission denied
```

```
chmod +x test.sh
./test.sh
```

```
Script executed successfully!
```

Instructor explains: Without execute permission, even a valid script cannot run. This is a key defense — malware needs execute permission.

Demo 7: Piping and Redirection

```
# Pipe: send output of one command to another
ls /etc | head -5
```

```
adduser.conf
alternatives
apt
bash.bashrc
ca-certificates
```

```
# Count files in /etc
ls /etc | wc -l
```

```
187
```

```
# Search within output
ls /etc | grep "pass"
```

```
passwd
```

```
# Save command output to a file  
date > report.txt  
whoami >> report.txt  
hostname >> report.txt  
cat report.txt
```

```
Wed Jan 15 11:00:00 EST 2025  
kali  
kali
```

```
# Chain commands with &&  
mkdir testdir && cd testdir && pwd
```

```
/home/kali/mylab/testdir
```

```
cd ~/mylab
```

Demo 8: Getting Help

```
# Manual pages (press q to quit)  
man ls  
  
# Quick help  
ls --help  
  
# What does a command do?  
whatis grep
```

```
grep (1) - print lines that match patterns
```

```
# Where is a command located?  
which ls  
which nmap  
which python3
```

```
/usr/bin/ls  
/usr/bin/nmap  
/usr/bin/python3
```

```
# Type of a command  
type ls
```

```
type cd
```

```
ls is aliased to 'ls --color=auto'  
cd is a shell builtin
```

Demo 9: Useful Shortcuts and Tricks

```
# Tab completion -- type partial command and press Tab  
# Type: cd Des[TAB]  
cd Desktop/  
  
cd ~  
  
# Up arrow -- recall previous commands  
# Press Up to cycle through command history  
  
# Ctrl+C -- cancel/stop a running command  
ping google.com  
# Press Ctrl+C to stop  
  
# Ctrl+L -- clear screen (same as typing 'clear')  
  
# Ctrl+A -- move cursor to beginning of line  
# Ctrl+E -- move cursor to end of line  
  
# !! -- repeat the last command  
echo "Hello"  
!!
```

```
Hello  
Hello
```

```
# Run last command with sudo  
apt update  
sudo !!  
  
# Wildcard: * matches anything  
touch file1.txt file2.txt file3.txt image.png  
ls *.txt
```

```
file1.txt  file2.txt  file3.txt
```

```
ls file?.txt
```

```
file1.txt  file2.txt  file3.txt
```

```
# Clean up
rm file1.txt file2.txt file3.txt image.png
```

Demo 10: Verifying Lab Tools Are Installed

```
# Check tools we will use this semester
echo "--- Tool Verification ---"

which nmap && echo "nmap found" || echo "nmap NOT found"
which wireshark && echo "wireshark found" || echo "wireshark
NOT found"
which openssl && echo "openssl found" || echo "openssl NOT
found"
which john && echo "john found" || echo "john NOT found"
which python3 && echo "python3 found" || echo "python3 NOT
found"
which curl && echo "curl found" || echo "curl NOT found"
which wget && echo "wget found" || echo "wget NOT found"
```

```
--- Tool Verification ---
/usr/bin/nmap
nmap found
/usr/bin/wireshark
wireshark found
/usr/bin/openssl
openssl found
/usr/bin/john
john found
/usr/bin/python3
python3 found
/usr/bin/curl
curl found
/usr/bin/wget
wget found
```

```
# Check versions
nmap --version | head -1
openssl version
python3 --version
```

Instructor: “All these tools are available. We will use each of them in different weeks throughout the semester.”

4 STUDENT TASKS

Complete ALL tasks independently. Demonstrate results to the instructor for marks.

Task 1: Environment Verification (10 marks)

Objective: Prove your VM is set up correctly and you can use basic commands.

Do the following and fill in the table:

uetblue!15 #	Command to Run	Your Output
1	whoami	
2	hostname	
3	uname -r	
4	pwd	
5	id	
6	ip addr show (write your VM's IP only)	
7	df -h / (write total and available space)	
8	free -h (write total RAM)	
9	vmware-toolbox-cmd -version	
10	date	

Then run these additional checks:

```
# Verify internet works
ping -c 2 google.com

# Verify you changed the default password
# (Just confirm to instructor -- don't show password)
```

 **Screenshot:** Terminal showing at least 5 of the above commands with outputs.

► **Show instructor:** Filled table + screenshot + confirm password was changed.

Task 2: Navigation Challenge (15 marks)

Objective: Navigate the Linux file system confidently.

Part A: Explore the System (5 marks)

Visit each directory below, run `pwd` and `ls | head -5`, and **record what you find**:

uetblue!15 Directory	What is stored here?	Name any 2 items you saw
/home/kali		
/etc		
/var/log		
/bin		
/tmp		

After exploring, return to your home directory:

```
cd ~  
pwd  
# Should show: /home/kali
```

Part B: Path Navigation (5 marks)

Perform each navigation and record the result of `pwd`:

```
# Start from home  
cd ~  
  
# Go to /etc  
cd /etc  
# pwd output: -----  
  
# Go to parent directory  
cd ..  
# pwd output: -----  
  
# Go to /var/log  
cd /var/log  
# pwd output: -----  
  
# Go directly home  
cd  
# pwd output: -----  
  
# Go to Desktop using relative path  
cd Desktop  
# pwd output: -----  
  
# Go back one directory  
cd ..  
# pwd output: -----
```

uetblue!15 Action	pwd Output
cd /etc	
cd .. from /etc	
cd /var/log	
cd (go home)	
cd Desktop	
cd .. from Desktop	

Part C: Finding Things (5 marks)

```
# How many files/directories are in /etc?
ls /etc | wc -l
# Answer: -----

# Find all files in /etc that have "network" in the name
find /etc -name "*network*" 2>/dev/null
# List what you found:
# 1. -----
# 2. -----

# What type of file is /bin/bash?
file /bin/bash
# Answer: -----

# Where is the 'ping' command located?
which ping
# Answer: -----
```

► **Show instructor:** All tables filled, navigation answers, and finding results.

Task 3: File and Directory Operations (20 marks)

Objective: Create, manipulate, and manage files and directories.

Part A: Build a Directory Structure (6 marks)

Create this **exact** structure in your home directory:

```
~/CSC201L/
|-- Lab01/
|   |-- exercises/
|   +-- submissions/
|-- Lab02/
|-- Lab03/
|-- Notes/
+-- Scripts/
```

Hint: Use `mkdir -p` to create nested directories in one command.

Verify by running:

```
ls -R ~/CSC201L
```

Record the output: _____

Part B: Create and Manipulate Files (8 marks)

```
cd ~/CSC201L/Lab01/exercises

# 1. Create a file with your info
echo "Name: [Your Full Name]" > student_info.txt
echo "Reg No: [Your Registration Number]" >> student_info.txt
echo "Date: $(date)" >> student_info.txt
echo "Course: CSC201L Information Security Lab" >> student_info
.txt

# 2. Verify
cat student_info.txt

# 3. Count the lines
wc -l student_info.txt
# How many lines? -----

# 4. Create a copy
cp student_info.txt ~/CSC201L/Lab01/submissions/
student_info_submitted.txt

# 5. Verify the copy exists
ls ~/CSC201L/Lab01/submissions/

# 6. Create a second file
echo "Lab 1: Introduction to Linux Terminal" > topics.txt
echo "Lab 2: CIA Triad" >> topics.txt
echo "Lab 3: Access Control" >> topics.txt
echo "Lab 4: Classical Encryption" >> topics.txt
echo "Lab 5: Symmetric Encryption" >> topics.txt

# 7. Search for a specific topic
grep "Encryption" topics.txt
# Output: -----

# 8. How many topics mention "Lab"?
grep -c "Lab" topics.txt
# Answer: -----
```

Part C: Advanced File Operations (6 marks)

```
cd ~/CSC201L/Lab01/exercises

# 1. Combine two files into one
cat student_info.txt topics.txt > combined.txt
cat combined.txt

# 2. Sort the topics file
sort topics.txt

# 3. Show the file in reverse
tac topics.txt

# 4. Replace text (change "Lab" to "Week")
sed 's/Lab/Week/g' topics.txt
# Does the original file change? Check:
cat topics.txt
# Answer (yes/no): -----

# 5. Save the modified version
sed 's/Lab/Week/g' topics.txt > topics_weeks.txt
cat topics_weeks.txt

# 6. Compare original and modified
diff topics.txt topics_weeks.txt
```

 **Screenshot:** Output of `ls -R ~/CSC201L` showing your complete directory structure.

► **Show instructor:** Directory structure, file contents, grep results, and diff output.

Task 4: File Permissions Practice (20 marks)

Objective: Set and verify file permissions correctly.

Part A: Create and Secure Files (8 marks)

```
cd ~/CSC201L/Lab01/exercises

# Create three files
echo "TOP SECRET: Server root password is Xy9#mK2" >
  secret_passwords.txt
echo "MEMO: Team meeting moved to Friday" > team_memo.txt
echo "NOTICE: Campus will be closed on Jan 20" > public_notice.
  txt
```

Set these permissions:

uetblue!15 File	Who Can Access	Numeric	chmod Command
secret_passwords.txt	Owner: read+write ONLY	?	chmod ?
team_memo.txt	Owner: rw, Group: r, Others: none	?	chmod ?
public_notice.txt	Owner: rw, Group: r, Others: r	?	chmod ?

After setting permissions, verify:

```
ls -la secret_passwords.txt team_memo.txt public_notice.txt
```

Fill in the results:

uetblue!15 File	Permission String from ls -la	Correct? (Y/N)
secret_passwords.txt		
team_memo.txt		
public_notice.txt		

Part B: Permission Experiments (8 marks)

Perform each experiment and record **exactly** what happens:

Experiment 1: No permissions at all

```
chmod 000 secret_passwords.txt
cat secret_passwords.txt
```

Record error message: _____

Experiment 2: Write-only (no read)

```
chmod 200 secret_passwords.txt
cat secret_passwords.txt
```

Record what happened: _____

```
echo "Can I write without reading?" >> secret_passwords.txt
```

Did the write work? (yes/no): _____

Experiment 3: Execute permission on a script

```
# Create a script
echo '#!/bin/bash' > greet.sh
```

```
echo 'echo "Hello $(whoami), today is $(date)"' >> greet.sh

# Try without execute permission
./greet.sh
```

Record error: _____

```
# Add execute permission
chmod 700 greet.sh
./greet.sh
```

Record output: _____

Experiment 4: Restore and verify

```
chmod 600 secret_passwords.txt
cat secret_passwords.txt
```

Can you read it now? (yes/no): _____

Part C: Permission Questions (4 marks)

Q4a. You saw that `chmod 200` lets you write but not read. Why would this ever be useful? (*Hint: think about a system log file that programs write to but shouldn't read back*)

Q4b. Your web server needs to serve a file to the public. The admin should be able to edit it. Group members should read it. What permission number would you use? Show your calculation:

```
Owner:  r(4) + w(2) + ?(?) = __
Group:  r(4) = __
Others: r(4) = __
Total:  ___
```

 **Screenshot:** `ls -la` showing all three files with correct permissions.

► **Show instructor:** Permission table, all 4 experiment results, and question answers.

Task 5: Piping and Text Processing (15 marks)

Objective: Chain commands together using pipes and process text.

Part A: Pipe Basics (5 marks)

```
# 1. Count how many items are in your home directory
ls ~ | wc -l
# Answer: -----

# 2. List only directories in /etc (lines starting with 'd')
ls -la /etc | grep "^d" | head -5
# Write first 3 directories you see:
# 1. -----
# 2. -----
# 3. -----

# 3. Find all .txt files in your CSC201L directory
find ~/CSC201L -name "*.txt"
# How many .txt files? -----

# 4. Show your command history and count total commands
history | wc -l
# Total commands in history: -----

# 5. Find commands in history that used 'chmod'
history | grep "chmod"
# How many times did you use chmod? -----
```

Part B: Text Processing (5 marks)

```
cd ~/CSC201L/Lab01/exercises

# Create a sample data file
echo "Ahmed:CS:3.8:Lahore" > students.txt
echo "Sara:IT:3.5:Karachi" >> students.txt
echo "Ali:CS:3.2:Islamabad" >> students.txt
echo "Fatima:SE:3.9:Lahore" >> students.txt
echo "Hassan:CS:3.1:Peshawar" >> students.txt
echo "Ayesha:IT:3.7:Lahore" >> students.txt

# 1. Show only CS students
grep "CS" students.txt
# How many CS students? -----

# 2. Show only students from Lahore
grep "Lahore" students.txt
# How many from Lahore? -----

# 3. Sort students alphabetically
sort students.txt

# 4. Sort by GPA (3rd field, numeric, reverse order)
sort -t: -k3 -rn students.txt
# Who has the highest GPA? -----
```

```
# 5. Extract only names (first field)
cut -d: -f1 students.txt
```

Part C: Output Redirection (5 marks)

```
cd ~/CSC201L/Lab01/exercises

# 1. Save system info to a report
echo "=== SYSTEM REPORT ===" > system_report.txt
echo "Date: $(date)" >> system_report.txt
echo "User: $(whoami)" >> system_report.txt
echo "Hostname: $(hostname)" >> system_report.txt
echo "Kernel: $(uname -r)" >> system_report.txt
echo "IP: $(ip -4 addr show | grep inet | grep -v 127 | awk '{print $2}')" >> system_report.txt
echo "Disk: $(df -h / | tail -1 | awk '{print $4}')" free" >>
    system_report.txt
echo "RAM: $(free -h | grep Mem | awk '{print $4}')" free" >>
    system_report.txt
echo "=== END REPORT ===" >> system_report.txt

# View the report
cat system_report.txt

# 2. Count lines in the report
wc -l system_report.txt
# Answer: -----

# 3. Search for specific info in the report
grep "IP" system_report.txt
# Output: -----

# 4. Copy report to submissions folder
cp system_report.txt ~/CSC201L/Lab01/submissions/

# 5. Verify
ls ~/CSC201L/Lab01/submissions/
```

 **Screenshot:** Contents of your `system_report.txt`

➤ **Show instructor:** Pipe command results, text processing outputs, and system report.

Task 6: Write a Bash Script (20 marks)

Objective: Create a script that gathers system information automatically.

Requirements

Write a script called `lab_setup_check.sh` in `~/CSC201L/Scripts/` that does **ALL** of the following:

uetblue!15 #	What the Script Must Do
1	Print a header with your name and registration number
2	Print the current date and time
3	Print the username and hostname
4	Print the kernel version
5	Print the VM's IP address
6	Print disk space available on /
7	Print total and free RAM
8	List all <code>.txt</code> files in <code>~/CSC201L/Lab01/exercises/</code>
9	Check if these tools are installed: <code>nmap</code> , <code>openssl</code> , <code>python3</code> , <code>wireshark</code> , <code>john</code> — print <code>INSTALLED</code> or <code>NOT FOUND</code> for each
10	Print “Lab setup verification complete!” at the end

Additional Requirements

- Script must start with `#!/bin/bash`
- Script must have at least **5 comments** explaining sections
- Script must have permission 700 (only owner can read/write/execute)

Hints

```
# To check if a tool is installed:
which nmap > /dev/null 2>&1 && echo "nmap: INSTALLED" || echo "
nmap: NOT FOUND"

# To get just the IP address:
ip -4 addr show | grep inet | grep -v 127.0.0.1 | awk '{print
$2}'

# To get free disk space:
df -h / | tail -1 | awk '{print $4}'

# To list .txt files in a specific directory:
ls ~/CSC201L/Lab01/exercises/*.txt 2>/dev/null
```

Creating the Script

```
# Create the script
nano ~/CSC201L/Scripts/lab_setup_check.sh

# Write your script (use the requirements table above)
# Save: Ctrl+X -> Y -> Enter

# Set permissions
chmod 700 ~/CSC201L/Scripts/lab_setup_check.sh

# Run it
cd ~/CSC201L/Scripts/
./lab_setup_check.sh

# Save output to a file
./lab_setup_check.sh > ~/CSC201L/Lab01/submissions/setup_report
.txt
```

Verification

```
# Show your script code
echo "=== SCRIPT CODE ==="
cat ~/CSC201L/Scripts/lab_setup_check.sh

# Show script permissions
echo ""
echo "=== PERMISSIONS ==="
ls -la ~/CSC201L/Scripts/lab_setup_check.sh

# Show script output
echo ""
echo "=== SCRIPT OUTPUT ==="
./lab_setup_check.sh

# Show saved report
echo ""
echo "=== SAVED REPORT ==="
cat ~/CSC201L/Lab01/submissions/setup_report.txt
```

 Take TWO screenshots:

1. Your script code (cat lab_setup_check.sh)
2. The script running and showing output

► Show instructor:

- Script code displayed with cat
- Script running without errors
- ls -la showing rwx--- permissions

- Saved report in submissions folder

5 EVALUATION CHECKLIST

For Instructor Use

uetblue!20 #	Task	Max	Obtained	✓
1	Environment Verification: VM running, all 10 fields filled, password changed, screenshot	/10		☐
2	Navigation Challenge: System explored, all paths correct, files found	/15		☐
3	File/Directory Operations: Correct structure, files created with content, grep/diff/sed used	/20		☐
4	File Permissions: Correct chmod values, all 4 experiments recorded, questions answered	/20		☐
5	Piping & Text Processing: All pipe commands correct, text processing done, report created	/15		☐
6	Bash Script: All 10 requirements met, 5+ comments, runs clean, permissions 700, output saved	/20		☐
uetblue!10	TOTAL	/100		

5.1 Detailed Rubrics

Task 2 — Navigation (15 marks)

uetblue!15 Part	Full Marks	Half Marks	Zero
Part A: Exploration (5)	All 5 directories explored correctly	3 correct	Fewer than 3
Part B: Navigation (5)	All 6 pwd outputs correct	4 correct	Fewer than 4
Part C: Finding (5)	All answers correct	2–3 correct	Fewer than 2

Task 4 — Permissions (20 marks)

uetblue!15 Part	Full Marks	Half Marks	Zero
Part A: Set permissions (8)	All 3 files correct	2 correct	0–1 correct
Part B: Experiments (8)	All 4 experiments recorded accurately	2–3 recorded	Fewer than 2
Part C: Questions (4)	Both answers correct with reasoning	1 correct	None correct

Task 6 — Script (20 marks)

uetblue!15 Criterion	Marks
Script executes without errors	/4
All 10 requirements present in output	/10 (1 each)
At least 5 meaningful comments	/2
Correct permissions (700)	/2
Output saved to submissions folder	/2

6 COMMON PROBLEMS AND FIXES

Problem	Fix
VM won't start: "VT-x disabled"	Restart PC → Enter BIOS → Enable Intel VT-x or AMD-V → Save → Reboot
No internet in VM	VM → Settings → Network → NAT → Check "Connect at power on"
Tiny screen resolution	Run: <code>sudo apt install open-vm-tools-desktop</code> → Reboot
Can't copy-paste host↔VM	Install VMware Tools → VM → Settings → Options → Guest Isolation → Enable both
"Permission denied"	Add <code>sudo</code> before the command
"command not found"	Run: <code>sudo apt install <package-name></code>
Terminal seems frozen	Press <code>Ctrl + C</code> to cancel
Forgot your new password	Restore snapshot from before you changed it
Mouse stuck in VM	Press <code>Ctrl + Alt</code> to release
Black screen on boot	VM → Settings → Display → Uncheck "3D Acceleration"
How to exit nano	<code>Ctrl + X</code> → <code>Y</code> → <code>Enter</code>
How to exit man/less	Press <code>q</code>
How to exit top	Press <code>q</code>

7 NEXT WEEK PREPARATION

Week 2: Demonstrating Confidentiality, Integrity, and Availability (CIA Triad) through file-based exercises and simulated attacks.

To Prepare:

- ✓ Make sure your VM works with internet access
- ✓ Take a snapshot of your current working state
- ✓ Practice the commands from today until you are comfortable
- Read briefly about the CIA Triad (Confidentiality, Integrity, Availability)

End of Week 1 Lab Manual

CSC201L — Information Security Lab

University of Engineering and Technology, Lahore
